

Specific for KL

$$\text{Pinsker} \\ \eta_\gamma = 1/(2X_\gamma)$$

Primal bound helper

$$\text{Prop. ??} \\ X_\gamma = \frac{\|b\|_1 U_\gamma}{\gamma} + \\ de^{-C_{\min}/\gamma}$$

$$C_i \geq C_{\min}$$

$$\|x^{(k)}\|_\infty \leq X_\gamma$$

$$\|u^{(k)}\|_\infty \leq \\ U_\gamma$$

$$D_\phi \leq \eta_\gamma |\cdot|^2$$

$$|A|_{1 \rightarrow 1} \text{ bound}$$

Thm ??

Thm ??

$$\Delta_k = O(\varepsilon) \\ k = O(M_\gamma X_0 / \varepsilon^2)$$

$$\Delta_k = O(M_\gamma / \gamma k) \\ M_\gamma := 2U_\gamma^2 |A|_{1 \rightarrow 1}^2 / \eta_\gamma$$

Main results

Dual bound helper

$$|\log(x_\gamma/z)|_\infty \leq H_\gamma$$

$$\text{Prop. ??} \\ U_\gamma = \\ 2\kappa(|C|_\infty + \gamma H_\gamma)$$

Ψ
non-expansive

$\kappa(A_1, A_2)$
def. (??)

Ψ monotone

Prop. ??

$$\Psi(x+1) = \Psi(x) + 1$$

Prop. ??

Prop. ??

Specific
for W_1

$$C_{\min} = \\ W_{\min}$$

$$|A|_{1 \rightarrow 1} = 1$$

$$\text{Prop. ??} \\ H_\gamma = \log(\cdots / W_{\min}) + \\ 2W_{\max}/\gamma$$

$$\text{Prop. ??} \\ \kappa = 2 \text{diam}(E)$$

$$\Sigma = \\ (\text{Id}, -\text{Id})$$

$$\tau = +1$$

Σ -signature

τ -balance

Non-expansiveness helpers